From pipeline readings to an evidence-linked investigation.

Subsea Lens monitors pressure measurements and uses three AI specialist roles to investigate anomalies. This deck explains what the system does, how the implementation works and what its recorded results establish. No speaker or running application is needed.

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Pages 1-4 give the overview. Continue in order for the engineering explanation, or choose a topic.

Agent: a model-driven role with specific instructions and permitted tools.

LLM: a large language model that interprets supplied evidence and proposes reads or answers.

Reading: use Previous/Next or Left/Right. Space and Page Down scroll normally. Read all shows the complete document.

An alert says something changed. The engineer needs to know what to do.

A pressure change can indicate a pipeline problem, an operating change or unreliable instrumentation. Several can occur together. Listing anomalous sensors does not separate these explanations or establish an inspection priority.

Three questions drive the investigation

01 **What is happening?**
Distinguish a possible leak pattern from sensor, data and operating explanations.

02 **Where should inspection focus?**
Identify the affected pipeline section and a defensible inspection interval.

03 **Why, and what needs checking?**
Connect the recommendation to readings and records, including uncertainty and next checks.

Why this is more than an alarm screen

The engineer must reconcile neighboring readings, paired instruments, network layout and records from the same time. A faulty measurement can make a local alert look like a physical event; a shared data path can make several alerts look like independent confirmation.

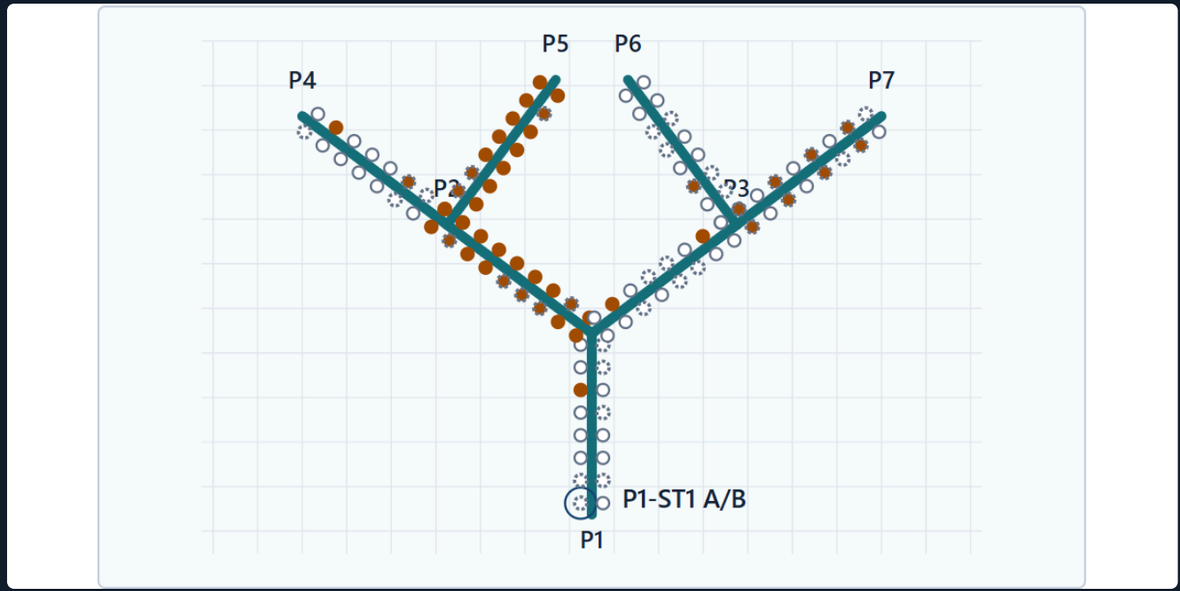
Subsea Lens organizes this investigation. Numerical monitoring identifies changes. The agents interpret the available evidence and prepare a reviewed, read-only handoff. The engineer retains the decision on inspection and deployment.

> The product's unit of value is an understandable investigation, not a larger collection of alerts.

Context: Executive brief.

One minute of data.
Sixty-nine flagged instruments.

The S7 demonstration contains seven pipeline sections and 112 pressure instruments. In the 60-second slice from source time 50 to 110 seconds, 69 instruments are flagged. The image below is the sensor-only view of that slice.



Sensor-only monitoring shows where measurements need attention. This is the actual dashboard crop, not a diagram of a model's reasoning.

What the engineer can see

The network view exposes the distribution of pressure alerts. Individual sensors can be inspected and compared with peers in the Live panel.

What the alerts do not answer

The engineer still has to decide which changes share a physical explanation, which readings are unreliable and where inspection should focus. **Sixty-nine flagged instruments do not mean 69 hardware failures or 69 leaks.**

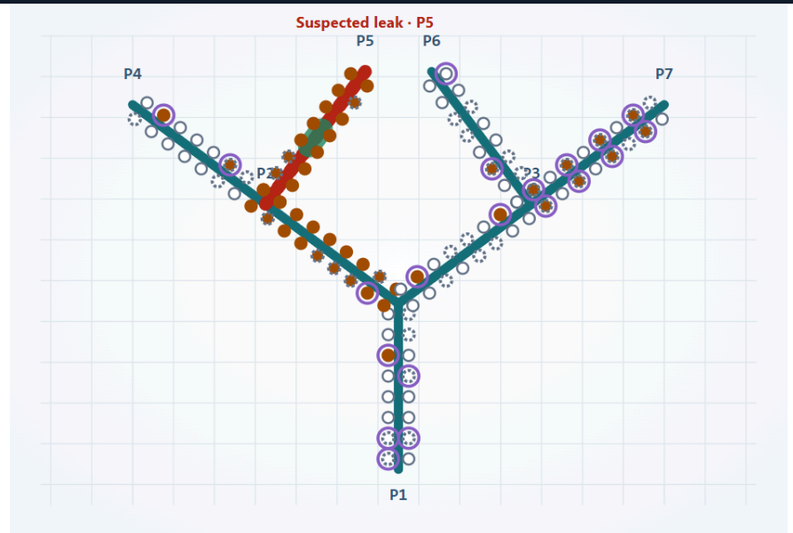
The next page uses matching measurements and detector alerts. It adds the saved agent investigation, rather than changing the input to produce a cleaner picture.

> A spatial alert display is a useful starting point. Interpretation is the missing handoff.

Capture and matching input hashes: brief-dashboard-comparison.json.

The agents turn that evidence into a connected inspection recommendation.

Sensor Quality checks reliability. Event Diagnosis compares pressure patterns and records. Operator Response assembles the explanation. The map below represents their report's findings for the same S7 slice.



Red: suspected P5 leak Green: inspection priority Purple: sensor/data concerns

The green interval is 700-900 m from P5's start, between P5-ST4 and P5-ST5. Colors encode the report's different findings, not three competing estimates of one thing.

WHAT THIS INVESTIGATION ADDS

Question 1 / Interpretation. A possible P5 leak pattern is separated from instrumentation problems elsewhere in the network.

Question 2 / Priority. The report identifies a P5 inspection interval, giving the engineer a concrete place to investigate.

Question 3 / Justification. Supporting readings and records, alternative explanations and further checks accompany the conclusion.

The engineer can review a connected explanation instead of assembling the investigation from individual alerts.

> The output is a supported investigation priority. Inspection and deployment remain engineering decisions.

Report quotation, interval and capture provenance: same-input comparison receipt. A separate grounding review precedes report acceptance.

Two connected paths: numerical monitoring and bounded investigation.

The application owns the data, execution rules and publication gates. The language model supplies interpretation inside those boundaries; it is not the monitoring engine or an unrestricted controller.

1 / OBSERVE

Raw observations

Arrival-gated readings, sensor metadata, network layout and a verified-normal baseline enter the measurement pipeline.

→

2 / MEASURE

Pinned window

Deterministic code computes quality and pressure summaries. Numerical warnings remain available without model access.

→

3 / INVESTIGATE

Three specialist roles

A bounded workflow gathers scoped evidence, distinguishes explanations and assembles an answer.

→

4 / PUBLISH

Reviewed handoff

Validation and source-grounding review gate the report. A separate extraction step maps its quoted findings.

Persistent application state: PostgreSQL stores observations, snapshots, investigation tasks, tool-read receipts and accepted outputs. Conversation checkpoints support resuming model work.

Human boundary: the dashboard shows warnings, reports, evidence and map overlays. This application does not operate pipeline equipment or dispatch inspection assets.

> An investigation is a governed extension of monitoring, not a replacement for it.

Engineering report; monitoring.py; runtime/workflow.py.

Preserve what arrived.
Standardize what can be compared.

The demonstration uses seeded synthetic observations. Ingest converts usable readings into a consistent representation while retaining the original payload and provenance. The source truth used for scoring is kept out of the agents' evidence.

Observations

A reading identifies its sensor and time, value and unit, arrival information and validity. Canonicalization uses UTC timestamps and pressure in bar so compatible readings can be compared.

Reference context

The site pack defines pipeline connectivity, instrument positions and paired stations. A separately verified-normal baseline establishes what ordinary pressure behavior looks like.

Quality and provenance

Missing or unusable readings are not filled with zero. Conflicting payloads with the same observation identity are quarantined. Original source information remains available for audit.

Scale in this example: S7 represents 11.2 km of pipeline: seven 1,600 m sections with 56 paired stations and 112 instruments.

Coordinate meaning: positions are local metres along the synthetic network, not GPS positions. The simulated signal patterns are not a calibrated hydraulic model of an operating asset.

> A model cannot repair a misleading data contract. Comparable measurements and explicit gaps come first.

Three clocks explain what the system knew, and when.

A replay can move 30 times faster than real time. Hosted model inference cannot. Without a pinned evidence window and an investigation hold, a valid report can appear beside a much later simulation frame and look incorrectly timed.

Event time	When the observation represents a measurement. Monitoring summarizes **60-second windows**, advancing by **5-second hops**, with a 5-second lateness allowance.
Arrival / source clock	When that observation is available to this replay. A window uses only observations received by its pinned cutoff. Later arrivals do not secretly appear in the earlier investigation.
Elapsed time	How long computation and hosted calls actually take. Assisted replay acquires configured post-alert context, then holds the source while the pinned investigation and result admission are handled.

What the hold fixes: ground truth, readings and agent output can be compared on the same simulated frame. **What it does not do:** make a two-minute hosted investigation finish in four real seconds at 30x replay. Report turnaround must still be measured separately.

> Time alignment is a correctness feature. It must not be presented as faster inference or instant detection.

monitoring.py / source advancement and investigation hold; Results / timing scopes.

Code measures the change. The agents assess its meaning.

Fresh agent-assisted investigations use measurement-only screening. Numerical tools expose pressure summaries, profiles, quality information and metadata; they do not supply a prewritten leak cause or inspection answer.

What deterministic code is good at

Compare readings with the normal baseline using robust median and median absolute deviation statistics. Count valid readings, inspect coverage and summarize differences consistently.

Compare paired instruments and nearby stations, using network connectivity rather than an arbitrary distance across unrelated branches. Correlations use paired-complete observations rather than invented samples.

What still requires interpretation

A large anomaly score indicates an unusual reading relative to a baseline. It is not a calibrated probability that a leak exists.

Several affected instruments may share a gateway or data path. Their agreement is useful context, but not automatically independent physical confirmation.

Fresh diagnosis policy: the model must explain competing causes from measurements and records, not merely repeat a deterministic event classifier.

> The numerical layer makes the evidence repeatable. It deliberately leaves the causal judgment open.

registry.py / LLM_REGISTRY. Legacy numerical diagnosis tools remain for older tasks; they are excluded from the fresh diagnosis tool policy.

Every conclusion needs an exact version of its inputs.

A snapshot is the stored measurement window an investigation is allowed to use. It binds the site, time range, source cutoff and numerical context. The dashboard's continually changing latest window is not a substitute for that identity.

SOURCE

Received data

Original observations and their availability at a particular source cutoff.

→

SNAPSHOT

Window + versions

Pinned network pack, baseline and parameter versions, with input hashes.

→

INVESTIGATION

Root + handoffs

One parent investigation and compatible specialist tasks bound to that snapshot.

→

OUTPUT

Report + receipts

Accepted claims, original tool reads and the sources used to check them.

Late evidence creates a new revision. It does not silently rewrite the old report's inputs. A resumed task must retain compatible evidence and policy pins.

Why hashes matter: two screenshots can look similar while using different evidence. The S7 comparison records matching measurement and alert hashes to establish that its inputs really match.

> Reproducibility means preserving evidence identity, not merely saving the final paragraph.

workflow.py / pinned submission and compatibility; S7 comparison hashes.

The roles divide the investigation. The application controls their handoffs.

An investigation root is the parent work item for one pinned question. Operator Response plans the specialist work and assembles the final answer. Quality precedes diagnosis so unreliable inputs can influence the physical interpretation.

FIRST / SENSOR QUALITY

Can these readings be trusted?

Inspect coverage, conflicting data, unusual patterns and peer relationships. Hand off specific quality concerns and limitations tied to the same snapshot.

THEN / EVENT DIAGNOSIS

Which explanation fits?

Use the quality handoff, pressure measurements, topology and selected records to distinguish leakage, instrumentation and operating explanations.

FINALLY / OPERATOR RESPONSE

What should the engineer review?

Assemble an evidence-linked answer: suspected cause, inspection priority, alternatives, uncertainty and further checks. Reuse earlier specialists only if their bindings are compatible.

Not a free-form agent conversation: work is executed through a persisted plan, role policies and checked dependencies. Specialists cannot silently switch the site or window.

Other model calls have different jobs: source-grounding review checks the answer; report-to-map extraction encodes accepted findings. Neither is a fourth diagnostic specialist.

> Three roles describe responsibility, not three autonomous controllers or exactly three model calls.

workflow.py; role policies and tools.

The model can request evidence. It cannot choose its own authority.

The application exposes role-specific tool schemas. A model proposes a tool name and structured arguments; the gateway validates them before executing a read. The owned Copilot runtime's native tools are disabled.

EXAMPLE / A PRESSURE-MEASUREMENT READ

Requested operation: `pressure.measurements`

Required bindings: site ID, snapshot ID, network-pack version, baseline version and parameter version.

Before reading: does this role permit the tool, does its argument schema match, and do all bindings equal the task's pinned evidence?

After reading: return structured evidence and retain a receipt identifying the operation, inputs and output.

Scope is enforced in code

A prompt saying "stay on this site" is not sufficient. A read with a different site, window or numerical version is rejected by the gateway.

Different roles have different capabilities

Quality, diagnosis and response receive appropriate tool sets. Fresh diagnosis can read measurements but cannot call the legacy deterministic leak-assessment or location-fitting tools.

Receipts expose public inputs and outputs for audit. They are not a claim to expose the model's private internal reasoning.

> A tool call is a checked request to the application, not permission to browse arbitrary data or execute arbitrary commands.

gateway.py / _authorize, _validate_scope and dispatch; registry.py.

Useful context must be found, read and judged for applicability.

Pressure patterns alone may not distinguish an event from an operating or data-path change. Agents can discover relevant stored records and read their contents. Retrieval is PostgreSQL lexical full-text search, not an unrestricted web search or a vector database.

Measurements

Establish what the instruments reported, where, and within which window. Quality concerns determine how strongly those readings support an explanation.

Operational records

Provide selected operating and historical context. Records are versioned and filtered for the incident time; a relevant search result must be read before it can support a citation.

Advisory knowledge

Provides reviewed demonstration guidance for interpreting patterns and planning checks. Applicability must be assessed; guidance is not itself a measurement or operational approval.

Example of the distinction: a common shift across instruments sharing one path can motivate a data-path hypothesis. A relevant record may support that explanation, but it does not prove that the pressure change elsewhere is also a data fault. The report must connect each claim to the evidence that actually supports it.

> Retrieval supplies candidate context. Reading, applicability and source-backed interpretation make it evidence.

Engineering report / knowledge and history; gateway.py / scoped reads.

A well-formed answer is not enough. The explanation must survive source review.

The application first checks structured results, scope and references. A separately prompted, tool-free model review then checks explanation units against the supplied source content. This review may use the same underlying model; it is not an independent human assessment.

DRAFT
Candidate answer
Claims, citations, inspection priority, alternatives and limitations from the response role.

→

VALIDATE
Deterministic checks
Check schema, scope, permitted claims, evidence references and dependency compatibility.

→

REVIEW
Source grounding
Compare explanation units with actual source content, rather than trusting citation presence alone.

→

RESOLVE
Accept or revise
Revise within a bounded loop. Reject or withhold when review fails, is inconclusive or exhausts limits.

Bounded repair: fresh interpretation review allows up to three revisions and stops on acceptance. Unchanged source-bound units can reuse valid checks.

No success-shaped fallback: failure does not substitute an unchecked report. A source-backed abstention can be accepted when evidence does not support a physical diagnosis.

> Grounding improves the connection between claims and evidence. It does not turn a model review into proof of physical truth.

workflow.py / interpretation review; grounding.py; validation.py.

The workflow can outlive one process or model request.

Investigations can take minutes and involve multiple external calls. PostgreSQL business state governs ownership, task dependencies and bounded work. LangGraph checkpoints preserve conversation progress. These responsibilities are related but deliberately separate.

Business execution

Root: the parent investigation, with a pinned snapshot and overall limits.

Tasks and leases: persisted specialist work and time-limited worker ownership prevent the conversation checkpoint from becoming the sole authority on what may execute.

Budgets and receipts: model/tool limits, deadlines, read provenance and outcome state remain attached to the investigation.

Conversation progress

LangGraph checkpoints retain messages and graph position so work can resume without treating a completed conversation as the entire application record.

Model and tool effects occur outside graph nodes. No database transaction is held open across hosted inference.

Resume and reuse still require compatible evidence and policy bindings; cached success is not automatically trusted.

> Durability means recoverable, owned and bounded execution, not just saving a chat transcript.

workflow.py; agent.py; checkpointer.py.

The map must represent the report, not invent another diagnosis.

After a report is accepted, a separate model-assisted extraction step identifies its stated pipeline, inspection interval and instrument concerns. Extracted findings must be linked to quotations from the accepted report and checked against the network.

S7 / AN ACTUAL ACCEPTED-REPORT QUOTATION

"Prioritize integrity-engineer inspection planning on P5 between P5-ST4 and P5-ST5, chainage 700–900 m."

Mapped result: pipeline P5; interval start 700 m; interval end 900 m; retained quotation and report path.

The mapping input did not include ground truth. The overlay represents what this report recommended.

Separate finding types stay separate

The suspected-leak indication, inspection interval and instrument/data concerns have different meanings. Their colors and geometry should not imply that every marker is a leak location.

Separate failures stay visible

An accepted report can exist even if map extraction fails. A mapper-only retry can repair the visualization without rerunning the whole investigation or erasing the original failure.

The full source quotation also calls the interval a pressure-deficit focus. It is an inspection recommendation, not an exact physical coordinate.

> A map is a source-linked representation of the accepted answer, not independent confirmation of it.

Report quotation and mapping scope: S7 comparison receipt.

Replay should preserve what was known at the original time.

The Live tab can show sensor-only monitoring, a saved agent-assisted recording or a new assisted investigation. The Evaluation tab compares recorded results with known simulated cases. These modes do not have the same inference or account requirements.

Sensor-only monitoring

Display measurements and numerical alerts without starting agents. Sensor inspection and peer comparison are available in the same Live panel.

No Copilot account is needed.

Recorded assistance

Read saved reports and mappings; reconstruct measurements at the original arrival cutoff. Do not backfill future observations or make new inference calls.

No Copilot account is needed when the recordings are available locally.

New assistance

Admit fresh scoped work, run the specialists, review the explanation and map accepted findings. Accelerated playback still waits at the investigation hold.

Copilot sign-in and model access are required.

What ships: the bundled Detection evaluation can be explored in a clean checkout. The full S1-S7 Live recording library and tracing services are not bundled; a developer machine's saved library must not be assumed to exist for another reader.

> Saved playback is an inspectable execution record, not a fresh live test disguised as a recording.

README / modes, prerequisites and walkthrough; Engineering report / recording reads.

A small application stack, with explicit model and data boundaries.

LAYER	IMPLEMENTATION	RESPONSIBILITY
Browser	TypeScript, esbuild	Live monitoring, inline inspection/comparison, reports, maps and Evaluation.
Application	Python, FastAPI, Pydantic, NumPy	Typed API contracts, ingest, numerical analysis and execution orchestration.
Persistence	PostgreSQL + LangGraph checkpoints	Business state, observations, evidence, receipts and recoverable conversation progress.
Model access	Owned persistent Copilot CLI SDK runtime	Hosted inference, isolated session per inference, native tools disabled; application-mediated evidence tools.
Observability	Optional OpenTelemetry, Langfuse / Jaeger	Trace public execution inputs, outputs, calls and failures where services are configured.

Data boundaries: site/tenant row-level security in PostgreSQL and role-scoped tool authorization constrain reads. Local operator login and request protection guard the application interface.

Deployment boundary: model inference is hosted, so this is not a local-only AI system. The local demo's authentication is not a complete enterprise identity, high-availability or operational approval system.

> The application owns the evidence gateway. The model provider owns hosted inference.

Engineering report / architecture and deployment; Copilot SDK adapter.

The controls add work.
They also make failures inspectable.

CHOICE	WHAT IT BUYS	WHAT IT COSTS
Numerical measurement before model interpretation	Consistent arithmetic and warnings that do not depend on AI availability.	Baseline, quality rules and source contracts need explicit maintenance.
Ordered specialist handoffs	Quality limitations reach diagnosis; the response integrates checked dependencies.	Sequential work lengthens the critical path compared with one unconstrained call.
Pinned snapshots and controlled tools	Reproducible, scoped evidence rather than a moving latest-data query.	Versioning, compatibility checks and bounded retrieval add implementation complexity.
Validation plus semantic grounding	Invalid structure and unsupported interpretation have separate publication gates.	Review adds model calls and latency, and can still make semantic mistakes.
Separate report-to-map extraction	Keep the diagnosis intact when visualization needs repair; retain quoted provenance.	Map availability can lag report availability and needs its own failure state.

> The goal is not the fewest model calls. It is a useful handoff with bounded authority and auditable evidence.

Implemented mechanisms: Engineering report. Tradeoff statements are engineering rationale, not measured superiority over another system.

Separate the answer from the answer key. Then score predefined checkpoints.

The held-out cohort uses three simulated extended-network recordings, each with 112 instruments. Each recording is assessed at source seconds 60, 150 and 360: nine network/checkpoint snapshots in total. Code and prompts were frozen between development and this held-out seed.

What the system receives

Arrival-gated observations, normal references, network metadata and permitted records. The agents and report mapper do not receive the ground-truth leak source.

What the scorer compares

Leak-pattern recognition, affected pipeline, instrumentation concerns, named suspect instruments and issue types. Inspection intervals are checked for containment of the known simulated source.

What the nine snapshots contain

Three clean 60-second controls: no agent investigation is expected or called.

Six investigation snapshots: accepted reports and explanations are assessed; four contain a leak target.

Five no-leak snapshots: the inspection interval should be withheld. These include the three clean controls.

This scheduled-checkpoint design assesses diagnosis at those frames. It does not measure continuous production trigger delay.

> A known answer makes scoring possible. Keeping it outside the investigation makes the comparison meaningful.

Held-out report; Results / extended-network evaluation.

Read the counts behind the reported 100% scores.

Precision measures how many reported positives were correct; recall measures how many expected positives were found. Both were 100% in the groups below. The counts, sampling unit and retained limitations are essential to interpreting that result.

SCORED GROUP	TRUE POSITIVES	FALSE POSITIVES / NEGATIVES	INTERPRETATION
Leak recognition	4	0 / 0	Four leak-bearing network/checkpoint snapshots.
Affected pipeline	4	0 / 0	Correct branch for each of those four leak targets.
Instrumentation recognition	6	0 / 0	Instrumentation concerns recognized in six snapshots.
Suspect instruments	72	0 / 0	Instrument/checkpoint labels, not 72 independent physical faults.
Instrument issue types	80	0 / 0	Instrument/issue/checkpoint labels; an instrument can have multiple labels.

Other outcomes: source containment 4/4, with mean inspection width 400 m; correct interval withholding 5/5; accepted reports and external explanation-judge passes 6/6 each. Checkpoints share recordings, so they are not independent trials. Four leak successes have a Wilson 95% lower bound of about 51%.

> These are exact results for this recorded cohort, not a guarantee of perfect performance on new assets.

Held-out metrics and intervals. Stored-IO checks were 36/38: two mapper/judge input-hash mismatches are retained despite matching outputs; see storage receipt.

Reviewed reports in 2.5-2.8 minutes in these two controlled runs.

These runs used the persistent Copilot SDK runtime and the observed model gpt-6.1-sol. Timing ends when the reviewed investigation completes; the separate report-to-map callback is excluded. They are not the earlier S7 screenshot's 110-second checkpoint.

S7 / SOURCE +120 SECONDS

149.328 s

Review accepted the candidate

8 model calls, 45 tool calls. All specialists succeeded; 12 operational checks passed.

Root: `0bbd3468-f0a2-5fcd-aefe-42f8a3eefa13`

S7 / SOURCE +180 SECONDS

167.812 s

Revision, then acceptance

8 model calls, 36 tool calls. All specialists succeeded; 12 operational checks passed.

Root: `77e15881-2662-5937-8752-6097918bab8a`

Why the persistent runtime matters: an owned process can serve isolated inference sessions without repeatedly launching the CLI. Ordered handoffs, tool reads and review still determine much of the critical path.

What the evidence permits: observed turnaround for two roots, not a latency percentile or guarantee. Earlier held-out evaluation investigations took 290-483 s under their own execution conditions; cohorts must not be mixed into one speed claim.

> Measure numerical alert time, reviewed-report time and map-ready time separately. Faster replay does not shorten hosted inference.

Latency execution receipts; Results / timing conditions. Historical comparisons also changed concurrency and CLI version; they are not a transport-only experiment.

An accepted result can be an abstention. A repaired failure should still leave a record.

A later recording qualification added 57 fresh S1-S6 investigations across source checkpoints. This checks execution and report handling across the scenario library. It is not 57 independent successful leak diagnoses.

30

Supported reports

Physical or instrumentation findings were supported enough to publish a report. Supported reports all obtained completed mappings after explicit repair where needed.

27

Reviewed abstentions

Accepted reports explained why the available evidence did not support a diagnosis. Playback retained raw measurements without inventing diagnostic overlays.

57

Final grounding passes

The 57 investigations passed their final grounding reviews and 684 operational checks. That outcome includes review, repair and abstention, not only first-attempt success.

Concrete retained failure: one report-to-map extraction required an explicit mapper-only retry taking 10.531 seconds. The accepted report was retained, the map was repaired, and the original failure stayed in the receipt. This is different from rerunning until a clean success history can be shown.

> Operational success means a supported, inspectable outcome, including an honest abstention when appropriate.

Recording qualification receipt; Results / current library. Sanitized receipts are committed; the complete runtime recording library is not.

Explore the saved evidence first.
Generate new investigations when needed.

The README provides copy-and-paste Windows installation and startup commands. Local operator credentials sign into the dashboard; they are separate from Copilot authentication, which is only needed for new AI investigations.

01 **Open the bundled Detection evaluation.**

After local startup and operator login, choose Evaluation. Read a clean checkpoint, then an investigation checkpoint. Compare the source frame, ground truth, report and mapped findings. This path does not require Copilot.

02 **Inspect measurements in Live.**

Run sensor-only monitoring, select a flagged instrument and use its inspection or peer comparison in the same panel. This shows the evidence the numerical layer exposes without invoking agents.

03 **Explore recorded assistance if it is available locally.**

Replay a saved report and follow its citations. The full S1-S7 Live library is not part of a clean checkout; do not confuse it with the bundled Detection report.

04 **Optionally generate a fresh assisted investigation.**

Sign in to Copilot with access to gpt-6.1-sol, run the provider check described in the README and start assistance. Observe the source hold, role handoffs, review outcome and later map publication.

Setup and guided walkthrough

After startup: Evaluation / Live dashboard. These loopback links need your local server, not a public hosted service.

> A no-account evaluation path and a no-account monitoring path let readers explore before enabling hosted AI.

Trace the core behavior through five implementation boundaries.

START HERE	WHAT TO TRACE	QUESTION THE CODE SHOULD ANSWER
`monitoring.py`	Source advance, snapshot creation, alert grouping, investigation hold and admission.	Which observations were available, and which exact window was investigated?
`runtime/workflow.py`	Parent root, specialist plan, owned tasks, pinned context and result publication.	Who may work, what is reused and what happens after interruption?
`runtime/registry.py` `runtime/gateway.py`	Role tool policy, argument validation, exact evidence pins and read receipts.	Can a model read another scope or obtain a precomputed diagnosis?
`validation.py` `grounding.py`	Structural checks, source-bound explanation units and revision validation.	What prevents an invalid or unsupported answer from being accepted?
`llm/copilot_sdk.py`	Owned runtime, isolated inference sessions, transport errors and captured events.	Which work is application logic, and which work depends on hosted inference?

> For each boundary, follow the evidence identity and failure state as carefully as the successful return value.

Broader route and contracts: Engineering report. Raw evaluation receipts are linked on the preceding results pages.

A working investigation system. A clear boundary around its claims.

Subsea Lens demonstrates a complete route from received observations to numerical alerts, scoped specialist work, reviewed explanations and quoted map overlays. The recorded cohorts make that route inspectable without requiring a new inference run.

What is implemented

Arrival-aware evidence windows, measurement-only fresh diagnosis, role-scoped tools, durable orchestration, semantic grounding, explicit abstention, recorded playback and separate map repair.

What is measured

Checkpoint-level diagnostic results on a frozen simulated cohort, report-turnaround receipts for specific executions and operational qualification across the local scenario library. These are separate evidence sets with different units.

What deployment would still require

Asset-specific source integration, calibrated baselines and hydraulic validation. Broader independent data and explicit alert-delay, false-alarm and report-latency targets.

Operator studies to establish usability and actual time savings. Enterprise identity, deployment hardening, availability and an approved operational response process.

None of those are implied by a perfect score in nine simulated snapshots.

> The contribution is not "an LLM finds a leak." It is an evidence-bound, recoverable workflow that makes a complex investigation easier to review.